

DUAL-TONE SINGLE BELL For Two Doors

SURESH CHANDRA DWIVEDI

In modern homes, apartments, and offices, multiple entrance doors are common. Traditionally, each door requires a separate doorbell, which increases cost and can confuse residents. Hearing a bell without knowing whether it is the main entrance or the back door is a frequent inconvenience. To address this, a single bell device has been designed for two doors, using a common bell that produces two distinct sound patterns depending on which door's switch is pressed. Fig. 1 shows the device prototype without its transformer.

Circuit and working

Fig. 2 shows the circuit diagram of the dual-tone single bell device for

two doors. It uses a popular LM556 dual-timer IC, which has two 555 timers built into one package. The entire circuit is divided into three sections: the power supply section, the tone generation section, and the output section.

The power supply section is built around transformer X1 and LM7805 (IC1) and provides a regulated 5V DC supply. A step-down transformer (X1) reduces the mains 230V AC voltage to 15V AC, which is then rectified by a bridge rectifier (BR1). An electrolytic capacitor (C1) smooths the DC output, while another capacitor (C2) removes high-frequency noise. Finally, the LM7805 voltage regulator ensures a steady and ripple-free 5V DC, which

powers the LM556 timer IC and associated components.

The tone generation section, built around LM556 (IC2), produces two tones. This IC has two inbuilt timers in a single package. Both are configured in astable mode to generate oscillating signals. The first timer (IC2A) is activated when switch S1 is pressed at door 1. Its frequency is determined by resistors R2 and R3 and capacitor C4, producing a specific tone. Similarly, the second timer (IC2B) is triggered by switch S2 at door 2, and its frequency is set by resistors R5 and R6 and capacitor C6. Due to the difference in tones, one can easily distinguish which doorbell has been activated.

Tone 1 is set by resistors R2 and R3 and capacitor C4 around IC2A. Increasing C4 decreases the frequency to produce a lower-pitched tone, while decreasing it increases the frequency for a higher-pitched tone. Adjusting R2 also changes the tone inversely with frequency.

Tone 2 is set by resistors R4 and R5 and capacitor C6 around IC2B. Increasing C6 lowers the frequency

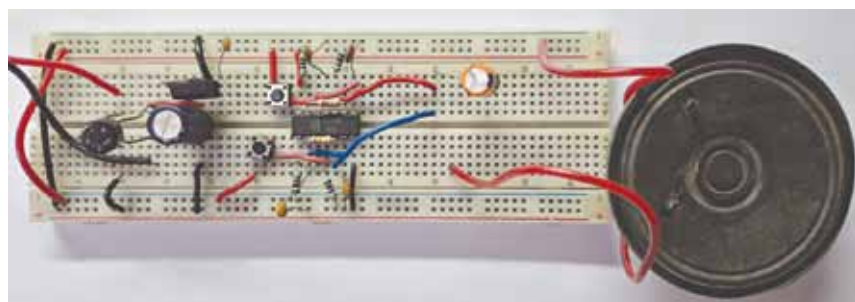


Fig. 1: Author's prototype

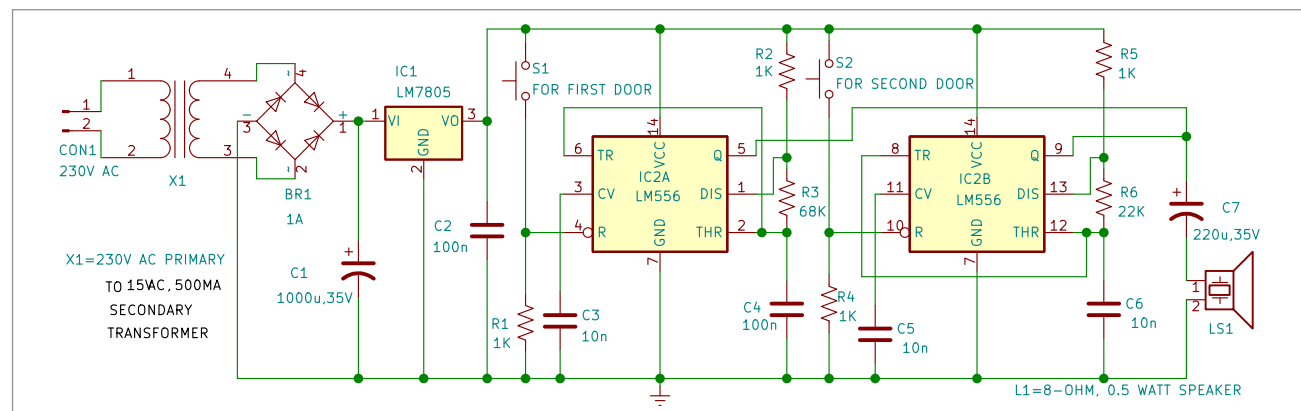


Fig. 2: Circuit diagram